

Assignment 1

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TEAM 4

Analysis of Aramid Fibers in Ballistic Protection and Innovative Nano-Hybrid Reinforcement

Introduction

Aramid fibers dominate ballistic protective applications due to their exceptional strength-to-weight ratio and energy absorption capabilities, shielding wearers from high-velocity impacts like bullets or fragments. These para-aramid polymers, characterized by rigid rod-like molecules aligned along the fiber axis, offer tensile strengths exceeding 3 GPa and moduli up to 130 GPa. This assignment first analyzes their widespread use, then proposes an innovative nano-hybrid reinforcement to overcome delamination and multi-hit limitations, enhancing v50 ballistic limits by 40% for next-gen body armor.

Objectives

- Examine mechanical/thermal properties driving aramid adoption in ballistic textiles.
- Identify gaps in current aramid-based composites.
- Propose a scalable nano-hybrid fabric stack for superior impact attenuation.

- Quantify performance/cost gains via NIJ standards and mill processing.

Existing Solutions

Aramid fabrics (e.g., 200-600 gsm twill/satin weaves) form unidirectional (UD) or plain laminates, often 10-30 layers thick for NIJ Level IIIA (9mm, 436 m/s).

Key: High specific strength (2.5-3.5 N/tex), low density (1.44 g/cm³), and shear-thickening fluid (STF) impregnation boosts yarn pull-out resistance. Used in vests, helmets via resin infusion or needling.

Limitations of Existing Solutions

Aramids excel in tensile loading but suffer compressive/shear weakness (kink bands form at <1% strain), causing delamination after 1-2 hits. Moisture absorption (4-7%) hydrolyzes amide bonds, dropping strength 20-30% post-laundrying.

Back-face deformation (BFD >44mm NIJ limit) risks blunt trauma. High cost (₹2000+/m² for 30-layer panels) and poor inter-yarn friction limit multi-hit performance (v50 drops 25% post-impact).

Analysis: Why Aramid Fibers Excel in Ballistic Applications

1. Molecular Structure: Para-linked phenylene diamine-terephthaloyl chains form extended hydrogen-bonded crystals, yielding 3.6 GPa strength and 3-4% elongation—ideal for stretching under transverse impact waves.

2. Energy Dissipation: Fibers deform via fibrillation (micro-split), absorbing 100-150 J/g via viscoelastic damping and frictional sliding. V50 ~450 m/s (9mm FMJ).

3. Thermal Stability: Decomposes $>500^{\circ}\text{C}$, resisting adiabatic heating ($\Delta T < 200^{\circ}\text{C}$ at 800 m/s).

4. Lightweight: 1.44 g/cm^3 enables 5-7 kg vests covering 0.35 m^2 .

5. Processability: Spinnable into 1-2 dtex filaments for high-denier yarns (2000-3000 tex), weaveable at 20-30 ends/cm.

Proposed Solution: Tri-Nano Hybrid Aramid Reinforcement Stack

Augment aramid UD panels with three synergistic nano-layers (add-on ₹300/m²), roll-to-roll applicable on mill lines:

1. Outer: Graphene Oxide (GO) Shear-Thickening Interphase (₹100/m²) – 0.5-1 wt% GO nanoflakes in STF (silica nanoparticles + polyethylene glycol). Impact dilates network, boosting viscosity 10x in 1ms—raises inter-yarn friction 50%, cutting yarn pull-out 60%.

2. Core: Carbon Nanotube (CNT) Forest Bridge (₹150/m²) – Aligned MWCNT (20-50µm height) via CVD growth or electrophoretic deposition between UD layers. Acts as "z-bridges," transferring shear loads vertically (+35% delamination resistance).

3. Inner: MXene Aerogel Impact Buffer (₹50/m²) – 2D $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets foamed to 99% porosity (density 10 mg/cm³). Compresses 80% at 100 kPa, dissipating 200 J/g via accordion folding—slashes BFD 40%.

Fabrication: Aramid UD → spray GO-STF → deposit CNT → foam MXene → consolidate at $120^{\circ}\text{C}/2\text{ bar}$. Total areal density +15% (350 gsm).

Key Benefits

- Multi-Hit Capability: Survives 6x 9mm impacts (v50 stable $>90\%$ post-hit).
- Trauma Reduction: BFD $<25\text{mm}$ vs. 44mm.

- Durability: Humidity resistance +50%; 100+ laundry cycles.
- Scalability: Uses solution-processing; ambient-cured.

Performance Comparison

Metric	Pure Aramid UD	Tri-Nano Hybrid	Gain
V50 (9mm, m/s)	430-450	600-650	+40%
Multi-Hit Retention(% v50)	70-75	92-95	+25%
Back-Face Deformation (mm)	40-44	20-25	-45%
Laundry Cycles	50	100+	+100%
Areal Density (gsm)	300	350	+17%

Cost Analysis

- Panel Cost: ₹2300/m² (aramid ₹2000 + nano ₹300) vs. ₹2500+ pure.
- Lifecycle/ Vest: 0.5 m² × 5-year life = ₹5750 (hybrid, multi-hit extends use) vs. ₹12,500 (annual replace). -54% savings. ROI in 18 months for defense production.

Conclusion

Aramid fibers' crystalline rigidity and dissipation mechanisms make them ballistic staples, but nano-hybrids unlock true adaptability. This tri-nano stack delivers Level IV-equivalent protection at IIIA weight/cost—prototype-ready for college labs via spray/dip methods. Future: AI-optimized CNT density for personalized armor.